UNITED STATES
NUCLEAR REGULATORY COMMISSION
REGION III
2056 WESTINGS AVENUE, SUITE 400
NAPERVILLE, IL 60563-2657

March 26, 2025

Mike Mlynarek
Site Vice President
Holtec Decommissioning
International, LLC
Palisades Nuclear Plant
27780 Blue Star Memorial Highway
Covert, MI 49043-9530

SUBJECT: PALISADES NUCLEAR PLANT – RESTART INSPECTION REPORT 05000255/2025001 AND 07200007/2024001

Dear Mike Mlynarek:

On February 28, 2025, the U.S. Nuclear Regulatory Commission (NRC) completed an inspection at the Palisades Nuclear Plant. The enclosed inspection report documents the inspection results, which the inspectors discussed on March 5, 2025, with yourself and other members of your staff.

One Severity Level IV violation without an associated finding is documented in this report. We are treating this violation as a non-cited violation (NCV) consistent with Section 2.3.2 of the Enforcement Policy.

If you contest the violation or the significance or severity of the violation documented in this inspection report, you should provide a response within 30 days of the date of this inspection report, with the basis for your denial, to the U.S. Nuclear Regulatory Commission, ATTN: Document Control Desk, Washington, DC 20555-0001; with copies to the Regional Administrator, Region III; the Director, Office of Enforcement; and the NRC Resident Inspector at the Palisades Nuclear Plant.

On June 13, 2022, Palisades ceased permanent power operations and subsequently removed all fuel from the reactor, as detailed in the letter from Entergy to the NRC, "Certifications of Permanent Cessation of Power Operations and Permanent Removal of Fuel from the Reactor Vessel," (ADAMS Accession No. ML22164A067). On September 28, 2023, Holtec Decommissioning International, LLC (Holtec) submitted a letter to the NRC requesting exemptions from certain portions of the requirements of Title 10 of the *Code of Federal Regulations* (10 CFR) 50.82(a)(2) to pursue the potential reauthorization of power operations at Palisades (ADAMS Accession No. ML23271A140). The NRC's initial acceptance to review Palisades' request for exemptions was documented in ADAMS at Accession No. ML23291A440 on November 3, 2023.

This letter, its enclosure, and your response (if any) will be made available for public inspection and copying at http://www.nrc.gov/reading-rm/adams.html and at the NRC Public Document Room in accordance with Title 10 of the *Code of Federal Regulations* 2.390, “Public Inspections, Exemptions, Requests for Withholding.”

Sincerely,

Signed by Nguyen, April
on 03/26/25

April M. Nguyen, Lead
Palisades Restart Team
Division of Operating Reactor Safety

Docket Nos. 05000255 and 07200007
License No. DPR-20

Enclosure:
As stated

cc w/ encl: Distribution via LISTSERV®

Letter to Mike Mlynarek from April Nguyen dated March 26, 2025.

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U.S. NUCLEAR REGULATORY COMMISSION
Inspection Report

Docket Numbers:	05000255 and 07200007
License Number:	DPR-20
Report Numbers:	05000255/2025001 and 07200007/2024001
Enterprise Identifiers:	I-2025-001-0078 and I-2024-001-0149
Licensee:	Holtec Decommissioning International, LLC
Facility:	Palisades Nuclear Plant
Location:	Covert, MI
Inspection Dates:	January 01, 2025 to February 28, 2025
Inspectors:	J. Austin, Senior Project Engineer, Region II B. Bergeon, Senior Operations Engineer D. Hoang, Senior Project Engineer B. Lin, Senior Health Physicist J. Mancuso, Senior Resident Inspector J. Nance, Operations Engineer T. Okamoto, Resident Inspector A. Sanchez, Senior Project Engineer, Region IV D. Sargis, Emergency Response Coordinator S. Schwind, Senior Resident Inspector, Callaway A. Shaikh, Senior Reactor Inspector C. Speer, Reactor Systems Engineer, NRR T. Steadham, Senior Resident Inspector, Browns Ferry D. Werkheiser, Senior Reactor Analyst, Region I J. Winslow, Senior Project Engineer, Restart Team
Approved By:	April M. Nguyen, Lead Palisades Restart Team Division of Operating Reactor Safety

Enclosure

SUMMARY

The U.S. Nuclear Regulatory Commission (NRC) performed inspections of the licensee’s activities to verify that the reactor coolant system boundary, reactor vessel head, and risk-significant piping system boundaries are appropriately monitored for degradation and that required repairs and replacements are appropriately examined and accepted; systems, structures, and components (SSCs) are appropriately inspected, maintained, and tested to ensure they can perform their required safety functions for potential restart; the licensed operator requalification training program and operator performance are appropriately performed; and independent spent fuel storage installation (ISFSI) cask loading activities are appropriately performed.

List of Findings and Violations

Failure to Correct an Inadequate Surveillance on the Spent Fuel Pool Overhead Crane			
Cornerstone	Severity	Cross-Cutting Aspect	Report Section
Not Applicable	Severity Level IV NCV 07200007/2024001-01 Open/Closed	Not Applicable	60855
The inspectors identified a severity level IV non-cited violation of Title 10 of the *Code of Federal Regulations* (10 CFR) Part 50, Appendix B, Criterion XVI, “Corrective Action,” for the licensee’s failure to establish measures to assure a condition adverse to quality was corrected. Specifically, the licensee did not take corrective action when they identified that the spent fuel pool overhead crane annual surveillance was performed without calibrated measuring and test equipment (M&TE). As a result, the safety-related work could not be validated, and the site could not maintain traceability of equipment used to perform measurements in the surveillance.			

Additional Tracking Items

None.

PLANT STATUS

The Palisades Nuclear Plant ceased power operations and subsequently removed all fuel from the reactor, as certified in a letter to the NRC on June 13, 2022. On September 28, 2023, Holtec Decommissioning International, LLC (Holtec) submitted a letter to the NRC requesting exemptions from certain portions of the requirements of 10 CFR 50.82(a)(2) to pursue the potential reauthorization of power operations. The NRC is conducting inspections to evaluate and assess the licensee's activities for potential restoration to an operational status.

The resident inspectors, as well as supplemental inspectors, continue to supply daily oversight of the restart activities occurring at the site.

INSPECTION SCOPES

Inspections were conducted using the appropriate portions of the referenced inspection procedures (IPs) in effect at the beginning of the inspection unless otherwise noted. Currently approved IPs with their attached revision histories are located on the public website at http://www.nrc.gov/reading-rm/doc-collections/insp-manual/inspection-procedure/index.html.

REACTOR SAFETY CORNERSTONES

71111.01 - Adverse Weather Protection

Impending Severe Weather (IP Section 03.02)

(1) The inspectors evaluated the adequacy of the overall preparations to protect risk-significant systems from impending winter weather due to a winter weather advisory during the week of January 20, 2025. The inspectors focused their reviews on winter weather preparations for the service water system, plant boiler heating system, and outdoor storage tanks.

71111.05 - Fire Protection

Fire Area Walkdown and Inspection (IP Section 03.01)

The inspectors evaluated the implementation of the fire protection program by conducting a walkdown and performing a review to verify program compliance, equipment functionality, material condition, and operational readiness of the following fire areas:

(1) Pre-fire Plan Area 13G: Spent Fuel Pool Heat Exchanger Room, 590′ elevation on February 19, 2025

71111.08P - Inservice Inspection Activities (PWR)

From January 7, 2025, to February 28, 2025, the inspectors observed and verified Holtec Palisades' nondestructive examination (NDE) and repair/replacement activities of the reactor pressure vessel head and Alloy 600 welds, conducted in accordance with the requirements of the ASME Code Section XI. The purpose of these inspections was to verify that the reactor coolant system boundary, reactor pressure vessel head, and risk-significant piping system boundaries are appropriately monitored for degradation. The inspectors also verified that the examination results were thoroughly reviewed and appropriately evaluated, and that required corrective actions to address identified indications and defects will be appropriately implemented. During this inspection period, the licensee completed NDE activities and evaluation of data for both the reactor pressure vessel head and Alloy 600 welds in the primary coolant system and began repair/replacement activities in both areas. Further NRC inspections will monitor repair/replacement activities as they are conducted.

The inspectors verified that the following nondestructive examination and repair/replacement activities were conducted appropriately per the ASME Code or required standard, and that any potential indications and defects were identified and evaluated at the proper thresholds:

<u>Nondestructive Examination and Welding Activities (IP Section 03.01)</u>

Alloy 600 Weld Inspection and Mitigation (January 7 through February 21, 2025)

- Surge Line Hot Leg Weld Overlay Fabrication
- Shutdown Cooling Hot Leg Weld Overlay Post Welding Ultrasonic Examination and Liquid Penetrant Examination
- Surge Line Hot Leg Weld Overlay Post Welding Ultrasonic Examination
- Safety Injection Cold Leg 1A and 2B Weld Overlay Fabrication

<u>Vessel Upper Head Penetration Inspection Activities (IP Section 03.02)</u>

Reactor Pressure Vessel Head Nozzle Repairs and Examinations

- All penetrations nozzles except nozzle #5 final nozzle machining and surface preparation (flapping to smooth out surface)
- All penetrations nozzles except nozzle #5 post-machining liquid penetrant examination
- Nozzle #5 inner diameter bore measurements

<u>71111.11Q - Licensed Operator Requalification Program and Licensed Operator Performance</u>

<u>Licensed Operator Requalification Program and Licensed Operator Performance</u>

(1) The inspectors observed quarterly licensed operator requalification training during an evaluated scenario on January 7, 2025.

<u>71111.13 - Maintenance Risk Assessments and Emergent Work Control</u>

<u>Risk Assessment and Management Sample (IP Section 03.01)</u>

The inspectors evaluated the accuracy and completeness of risk assessments for the following planned work activities to ensure configuration changes and appropriate work controls were addressed:

(1) Primary Coolant Pump ‘D’ lift activities during the work weeks of January 10 through January 21, 2025
(2) Dry Fuel Storage Cask lift activities on January 10, 2025
(3) Turbine Generator Rotor lift activities on January 25, 2025
(4) Low Pressure Turbine ‘B’ hood cover lift activities on February 4, 2025

OTHER BASELINE ACTIVITIES

71152 - Problem Identification and Resolution

System Return to Service

During the inspection period, the inspectors continued reviews of Phases I and II of the licensee’s system return to service (SRTS) plans. The SRTS plans document licensee activities to verify that the configuration and condition of Systems, Structures, and Components (SSCs) are consistent with the design and licensing bases to support potential restart. The inspectors reviewed Phases I and II of the SRTS plans, which include areas such as licensing and design-basis functions, system material conditions, reviewing open and deferred work prior to shutdown, and program applicability. The inspectors conducted these reviews using IP 71152, “Problem Identification and Resolution,” and IP 71111.21M, “Comprehensive Engineering Team Inspection (CETI),” as applicable.

The inspectors commenced reviews of the SRTS plans for the following systems:

- Control Rod Drive System
- Dry Fuel Storage
- Containment and Fuel Pool Cranes

The inspectors completed initial reviews of the SRTS plans for the following systems:

- Auxiliary Feedwater (AFW)
- AC Power Systems (4160 Volt and 2400 Volt)
- Service Water
- DC Power Systems (Vital and Non-vital)
- Emergency Diesel Generators (EDGs)
- Low Pressure Safety Injection and Shutdown Cooling

These systems were selected based on the importance of the SSCs during normal and emergency plant operating conditions, risk significance and defense-in-depth of the SSC’s specific functions and required work/modifications of the SSCs scheduled prior to potential restart.

The inspectors evaluated the initial SRTS plans for each system to identify if planned licensee activities will verify that the configuration and condition of the SSCs are consistent with the proposed operational design and licensing bases to support restart and verify readiness of the SSCs to support a return to service. The inspectors noted that the SRTS plans are intended to be living documents with a phased approach to SRTS. The initial plans were noted to be at a very high level and did not always provide specific details associated with inspection or testing plans. The inspectors noted that this specific level of detail is still in progress and, as the licensee progresses through the SRTS phases, the SRTS plans will undergo additional reviews by NRC inspector to ensure SSCs.

OTHER ACTIVITIES – TEMPORARY INSTRUCTIONS, INFREQUENT AND ABNORMAL

60855 - Operation of an ISFSI

Operation of an ISFSI (1 Sample)

(1) The inspectors evaluated the licensee's independent spent fuel storage installation (ISFSI) cask loading activities from December 16, 2024, through January 10, 2025. Specifically, the inspectors observed the following activities during the loading of multi-purpose canister (MPC) No. 426:

1. Fuel loading
2. Heavy load movement of loaded MPC out of the spent fuel pool
3. Canister lid welding
4. Nondestructive evaluation of canister welds
5. Radiological field surveys
6. Forced helium dehydration

The inspectors performed walkdowns of the ISFSI pads, walkdowns of the ISFSI haul path, and independent radiation surveys.

The inspectors evaluated the following:

1. Spent fuel selected for loading into dry cask storage during this loading campaign
2. Selected corrective action program documents
3. Selected 10 CFR 72.48 screenings and evaluations
4. Changes to the site's 10 CFR 72.212 report since the previous inspection

INSPECTION RESULTS

Failure to Correct an Inadequate Surveillance on the Spent Fuel Pool Overhead Crane			
Cornerstone	Severity	Cross-Cutting Aspect	Report Section
Not Applicable	Severity Level IV NCV 07200007/2024001-01 Open/Closed	Not Applicable	60855
The inspectors identified a severity level IV non-cited violation of Title 10 of the *Code of Federal Regulations* (10 CFR) Part 50, Appendix B, Criterion XVI, "Corrective Action," for the licensee's failure to establish measures to assure a condition adverse to quality was corrected. Specifically, the licensee did not take corrective action when they identified that the spent fuel pool overhead crane annual surveillance was performed without calibrated measuring and test equipment (M&TE). As a result, the safety-related work could not be validated, and the site could not maintain traceability of equipment used to perform measurements in the surveillance.			

Description: On May 13, 2024, the licensee performed work order 50086009-01-01-009 and procedure MSM-M-13C, “Spent Fuel Pool Crane L-3 Periodic Inspection.” Step 3.6.1 states specific M&TE to be used during the surveillance and lists a table to document the identification number and calibration due date for each piece of equipment. The licensee did not document any M&TE data for the ammeter, multimeter, or dial caliper. For the torque wrench, the licensee documented a torque wrench that was out of calibration. The work order was signed off as complete on May 13, 2024. On August 5, 2024, the licensee wrote corrective action document PAL-02834. The licensee identified the failure to document the use of calibrated M&TE. The licensee also identified that M&TE traceability was not being maintained in accordance with the site quality assurance program. The licensee made a corrective action to perform a work group evaluation to determine the cause of the issue. The licensee subsequently created an action to inform other work groups of the vulnerability of not using calibrated M&TE. However, the licensee did not create any actions to address the fact that the surveillance was not done in accordance with the licensee’s procedure and the site quality assurance program. The corrective actions were closed out and the issue report was closed on December 2, 2024. During the cask loading campaign beginning in December 2024, the inspectors reviewed the spent fuel pool crane periodic inspection and associated corrective action documents and identified that the crane inspection still did not have traceability of M&TE used to perform the measurements. The licensee stopped work with the crane and performed work order 00588078-01 to re-do the portions of the annual crane inspection that required the use of calibrated M&TE. The licensee identified that the crane emergency brake torque was higher than the allowed tolerance. The licensee adjusted the brake torque to within the allowed tolerance and documented the condition in the corrective action program. The inspectors noted that the licensee had sufficient time to create and complete a corrective action to restore the crane to full compliance because 4 months elapsed from the identification of the issue and the closure of the issue report. Additionally, the licensee performed a heavy lift of a loaded spent fuel canister following the closure of the issue report that identified the crane inspection was done without calibrated M&TE. It was determined that there was no impact to lifting the canister or to the structural integrity of the crane during this heavy lift. Corrective Actions: The licensee performed work order 00588078-01 “L-3 Return to Operability,” in order to re-do sections of the spent fuel crane annual surveillance that required calibrated M&TE. The licensee set the emergency brake torque to the correct tolerance. Corrective Action References: PAL-04948 and PAL-04960
Performance Assessment: None
Enforcement: Severity: The inspectors determined that the issue was more than minor because it was similar to example 4.c of Inspection Manual Chapter (IMC) 0612, Appendix E, “Examples of Minor Issues.” Specifically, the crane was used to perform a heavy lift of a loaded spent fuel canister and subsequent testing identified that the emergency brake torque measurement

was outside of tolerance. The inspectors determined that the issue was of very low safety significance because it did not affect the structural qualification of the crane and there were no actual consequences as a result of this violation. Therefore, the violation was determined to be a severity level (SL) IV non-cited violation (NCV).

Violation: Title 10 CFR Part 50, Appendix B, Criterion XVI, “Corrective Actions,” requires, in part, that measures shall be established to assure that conditions adverse to quality, such as failures, malfunctions, deficiencies, deviations, defective material and equipment, and nonconformances are promptly identified and corrected.

Contrary to the above, from August 5, 2024 to January 18, 2025, the licensee failed to establish measures to assure that conditions adverse to quality were promptly corrected. Specifically, the licensee identified that the safety-related L-3 spent fuel pool overhead crane annual surveillance was performed without calibrated measuring and test equipment. The licensee entered the issue into the corrective action program as PAL-02834. The licensee failed to correct or take corrective actions to address the improper M&TE usage for the surveillance and closed PAL-02834 on December 2, 2024.

Enforcement Action: This violation is being treated as a non-cited violation, consistent with Section 2.3.2 of the Enforcement Policy.

EXIT MEETINGS AND DEBRIEFS

The inspectors verified no proprietary information was retained or documented in this report.

- On March 5, 2025, the inspectors presented the inspection results to M. Mlynarek, Site Vice President, and other members of the licensee staff.
- On January 30, 2025, the inspectors presented the ISFSI cask loading inspection results to M. Mlynarek, Site Vice President, and other members of the licensee staff.

DOCUMENTS REVIEWED

Inspection Procedure	Type	Designation	Description or Title	Revision or Date
60855	ALARA Plans	EN-RP-105	ALARA Plan	20
	Corrective Action Documents	PAL-02834	Quality Assurance (QA) Identified Finding	08/05/2024
		PAL-04592	FME in SFP	12/16/2024
		PAL-04834	Loss of Power to Helium Dehydration (FHD) Skid	01/10/2025
	Corrective Action Documents Resulting from Inspection	PAL-04947	L-3 Spent Fuel Pool Overhead Crane	01/16/2025
		PAL-04948	Inadequate Disposition and Actions Related to IR PAL-02834	01/16/2025
		PAL-04949	NRC Identified - L-3 (Spent Fuel Pool Crane) Readings	01/16/2025
		PAL-04960	L-3 Crane Emergency Brake Torque Out of Tolerance	01/17/2025
	Engineering Changes	24-0464	Perform Modification to Change East Trolley Limit Switch on L-3	0
		24-0551	HPP-3226 Series Procedures Changes	0
		72.48 - 1677	Review of Design Change Package EC 93344 and All Holtec Reports Applicable to Palisades Nuclear Power Plant	0
	Miscellaneous	HI-2241072	Palisades VCC & Pad Inspection 2024	11/12/2024
	NDE Reports	HSP-507	MPC #426 Lid to Shell Welds + Hydro	01/07/2025
	Procedures	ENG-423	Fuel Selection for Holtec Hi-Storm FW MPC Storage System	11
		HPP-3226-0300	MPC Sealing, Drying, and Backfilling at Palisades	0
		HPP-3226-0400	MPC Stack-Up and Transfer at Palisade	0
		MS LT-MPC-Holtec	Helium Leak Test Report for Multi Purpose Canister (MPC) Vent and Drain Cover Plate	3665-03
	Radiation Surveys	2025-0101	Contamination Tech Spec Survey MPC #427	01/21/2025
	Work Orders	00588078-01	L-3, Return to Operability	01/18/2025
		50086009-01-01-0009	L-3, Inspect Fuel Pool Crane	05/13/2024
71111.01	Corrective Action Documents	PAL-05032	Frozen tube to service water compositor	01/21/2025
		PAL-05045	LIT-1336B, F-4B Bay Level West Side, Failed LOW	01/21/2025
		PAL-05056	Cooling Tower Pump House Temperature	01/22/2025

Inspection Procedure	Type	Designation	Description or Title	Revision or Date
		PAL-05070	Hydrogen Monitor C-164 Heat Trace Panel Contactor Failed	01/22/2025
	Procedures	SOP-15, Attachment 10	Frazil Ice - Information/Prevention/Mitigation	Revision 73
		SOP-23, Attachment 14	Actions When Outside Temperatures Are Less Than 20F	Revision 71
71111.05	Fire Plans	PFP	Pre-Fire Plan	Revision 6
	Procedures	FPIP-1	Fire Protection Plan, Organization, and Responsibilities	Revision 28
		FPIP-4	Fire Protection and Fire Protection Equipment	Revision 44
71111.08P	NDE Reports	HLSC 350-00-PT-001	Liquid Penetrant Examination of Hot Leg Shutdown Cooling Nozzle Weld Overlay	01/21/2025
		HLSC 360-00-UT-001	Ultrasonic Examination of Elbow to Safe End to Nozzle Overlay	01/22/2025
	Procedures	54-ISI-864-007	Manual Phased Array Procedure for Weld Overlaid Similar and Dissimilar Metal Welds	02/21/2019
		54-PT-200-026	Color Contrast Removable Liquid Penetrant Examination of Components	10/31/2023
		LMT-10-PAUT-019	Manual Phased Array Ultrasonic Examination of Dissimilar Metal Piping Welds	019
71111.11Q	Miscellaneous	PLSEG-LOR-25A-06, Performance Exam	Simulator Exercise Guide	0
71111.20	Procedures	MA-010	Material Handling Program	Revision 2
71152	Engineering Changes	EC 92577	Decommissioning Equipment Classification and FSAR Markups	0
	Procedures	ENG-105	System Return to Service Plans	1